Week 3: SQL Mastery with PGExercises

Joins, Aggregation, and Subqueries in Supabase

Why SQL Mastery Matters

- SQL = lingua franca of data analysis & DBA work
- Advanced queries enable reporting, monitoring, analytics

Recap from Week 2

- SELECT basics: columns, filtering, ordering
- INSERT, UPDATE, DELETE
- Intro to Supabase SQL editor

PGExercises Dataset

- Gym membership system dataset (members, facilities, bookings)
- Useful because: multiple related tables → joins & aggregates

INNER JOIN Basics

- Syntax: SELECT … FROM t1 JOIN t2 ON condition
- Example: Get member names + facility names they booked
- SQL:
  SELECT m.surname, f.name
  FROM cd.members m
  JOIN cd.bookings b ON m.memid=b.memid
  JOIN cd.facilities f ON b.facid=f.facid;

LEFT vs. INNER JOIN

- INNER JOIN → only matches
- LEFT JOIN → keep all from left even if no match
- Example: List all members, show bookings if exist

Aggregation (COUNT, SUM, AVG, MIN, MAX)

- GROUP BY usage
- Example: Count how many bookings per facility
- SQL:
  SELECT facid, COUNT(*) FROM cd.bookings GROUP BY facid;

HAVING Clause

- Filtering on aggregates
- Example: Facilities with > 1000 bookings

Subqueries (Single Row)

- Definition: query inside query
- Example: Members who joined after the avg join date
- SQL:
  SELECT firstname, surname
  FROM cd.members
  WHERE joindate > (SELECT AVG(joindate) FROM cd.members);

Subqueries (IN, EXISTS)

- Example: Members who booked tennis courts
- SQL:
```
SELECT firstname, surname
FROM cd.members
WHERE memid IN (
 SELECT memid FROM cd.bookings WHERE facid IN (0,1)
);
```

Combining Joins + Aggregates

- Real-world query: Top 5 members by total spending
- Shows multiple concepts in one query

Best Practices in SQL

- Write readable queries (indent & alias)
- Test subqueries separately
- Use EXPLAIN to check performance
- Verify assumptions with sample queries